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Anatomy & Physiology Workbook For Dummies®, 3rd Edition with Online Practice

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Introduction

Whether your aim is to become a physical therapist or a pharmacist, a doctor or an acupuncturist, a nutritionist or a personal trainer, a registered nurse or a paramedic, a parent or simply a healthy human being — your efforts have to be based on a good understanding of anatomy and physiology. But knowing that the knee bone connects to the thigh bone (or does it?) is just the tip of the iceberg. In *Anatomy & Physiology Workbook For Dummies*, 3rd Edition, you discover intricacies that will leave you agog with wonder. The human body is a miraculous biological machine capable of growing, interacting with the world, and even reproducing despite any number of environmental odds stacked against it. Understanding how the body's interlaced systems accomplish these feats requires a close look at everything from chemistry to structural mechanics.

Early anatomists relied on dissections to study the human body, which is why the Greek word *anatomia* means “to cut up or dissect.” Anatomical references have been found in Egypt dating back to 1600 BC, but it was the Greeks — Hippocrates, in particular — who first dissected bodies for medical study around 420 BC. That's why more than two millennia later we still use words based on Greek and Latin roots to identify anatomical structures.

That's also part of the reason so much of the study of anatomy and physiology feels like learning a foreign language. Truth be told, you are working with a foreign language, but it's the language of you and the one body you're ever going to have.

About This Book

This workbook isn't meant to replace a textbook, and it's certainly not meant to replace going to an actual anatomy and physiology class. It is designed as a supplement to your ongoing education and as a study aid in prepping for exams. That's why we give you insight into what your instructor most likely will emphasize as you move from one body system or structure to the next.

Your coursework might cover things in a different order than we've chosen for this book. We encourage you to take full advantage of the table of

contents and the index to find the material addressed in your class. Whatever you do, certainly don't feel obligated to go through this workbook in any particular order. However, please do answer the practice questions and check the answers at the end of each chapter because, in addition to answers, we clarify why the right answer is the right answer and why the other answers are incorrect; we also provide you with memory tools and other tips whenever possible.

Within this book, you may note that some web addresses break across two lines of text. If you're reading this book in print and want to visit one of these web pages, simply key in the web address exactly as it's noted in the text, pretending as though the line break doesn't exist. If you're reading this as an e-book, you've got it easy — just click the web address to be taken directly to the web page.

Foolish Assumptions

In writing *Anatomy & Physiology Workbook For Dummies*, 3rd Edition, we had to make some assumptions about you, the reader. If any of the following apply, this book's for you:

- » You're an advanced high school student or college student trying to puzzle out anatomy and physiology for the first time.
- » You're a student at any level who's returning to the topic after some time away, and you need some refreshing.
- » You're facing an anatomy and physiology exam and want a good study tool to ensure that you have a firm grasp of the topic.

Because this is a workbook, we had to limit our exposition of each and every topic so that we could include lots of practice questions to keep you guessing. (Believe us, we could go on forever about this anatomy and physiology stuff!) In leaving out some of the explanation of the topics covered in this book, we assume that you're not just looking to dabble in anatomy and physiology and therefore have access to at least one textbook on the subject.

Icons Used in This Book

Throughout this book, you'll find symbols in the margins that highlight critical ideas and information. Here's what they mean:



TIP

The Tip icon gives you juicy tidbits about how best to remember tricky terms or concepts in anatomy and physiology. It also highlights helpful strategies for fast translation and understanding.



REMEMBER

The Remember icon highlights key material that you should pay extra attention to in order to keep everything straight.



WARNING

This icon — otherwise known as the Warning icon — points out areas and topics where common pitfalls can lead you astray.



EXAMPLE

The Example icon marks questions for you to try your hand at. We give you the answer straightaway to get your juices flowing and your brain warmed up for more practice questions.

Beyond the Book

In addition to the material in the print or e-book you're reading right now, this product also comes with some access-anywhere goodies on the web. While it's important to study each anatomical system in detail, it's also helpful to know how to decipher unfamiliar anatomical terms the first time you see them. Check out the free Cheat Sheet by going to www.dummies.com and typing for “*Anatomy & Physiology Workbook For Dummies* cheat sheet” in the Search box.

You also get access to our online database of questions with even more practice for you. It contains an interactive quiz for each chapter, allowing you to hone your new knowledge even more!

To gain access to the online practice, all you have to do is register. Just follow these simple steps:

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2. **Go to Dummies.com and click *Activate Now*.**

3. **Find your product (*Anatomy & Physiology Workbook For Dummies*, 3rd Edition) and then follow the on-screen prompts to activate your PIN.**

Now you're ready to go! You can come back to the program as often as you want. Simply log in with the username and password you created during your initial login. No need to enter the access code a second time.

Where to Go from Here

If you purchased this book and you're already partway through an anatomy and physiology class, check the table of contents and zoom ahead to whichever segment your instructor is covering currently. When you have a few spare minutes, review the chapters that address topics your class already has covered. It's an excellent way to prep for a midterm or final exam.

If you haven't yet started an anatomy and physiology class, you have the freedom to start wherever you like (although we suggest that you begin with [Chapter 1](#)) and proceed onward and upward through the glorious machine that is the human body!

Part 1

The Building Blocks of the Body

IN THIS PART ...

Learn the language of anatomy and physiology.

Explore the basic building blocks and functions that make the parts of the body what they are. Dig into atoms, elements, chemical reactions, and metabolism.

Crack open the cell to see what's happening at life's most fundamental level. Find out about the cell membrane, the nucleus, organelles, proteins, and the cell life cycle.

Plunge into cell division, which has several phases: interphase, prophase, metaphase, anaphase, telophase, and cytokinesis.

Use histology to build all of the body's tissues — epithelial, connective, muscular, and nervous — from the inside out.

Chapter 1

The Language of Anatomy & Physiology

IN THIS CHAPTER

- » Learning to speak the language
 - » Casing the cavities
 - » Orienting yourself to the body
-

Human *anatomy* is the study of our bodies' structures while *physiology* is how they work. It makes sense, then, to learn the two in tandem. But before we can dive in to the body systems and their intricate structures, you must first learn to speak the language of the science.

Organization of the Body

As you know, the body is organized into systems, grouping together the organs that work together to achieve a common goal. To house all these organs, our body must create spaces to hold them. The body has two cavities that achieve this: the *dorsal cavity*, which holds the brain and spinal cord and the *ventral cavity* that holds everything else. The dorsal cavity splits into the *spinal cavity*, which holds the spinal cord, and the *cranial cavity* that houses the brain. The ventral cavity is split into the *thoracic cavity* and the *abdominopelvic cavity* by a large band of muscle called the *diaphragm*. Within the thoracic cavity are the right and left *pleural cavities*, which hold each lung, and the *mediastinum*. Within the mediastinum is the *pericardial cavity* which contains the heart. The abdominopelvic cavity divides into the *abdominal cavity* (with the stomach, liver, and intestines) and the *pelvic cavity* (with the bladder and reproductive organs), though there's no distinct barrier between the two.

In order to create these cavities within our bodies, we have membranes to

border the space. The *visceral membrane* lies atop of the organs, making direct contact with them. For example, the outermost layer of the heart is called the *visceral pericardium* and on the lungs it's the *visceral pleura*. The *parietal membrane* lies on the other side of the spaces or lining the cavity itself. So the lining of the abdominopelvic cavity is known as the *parietal peritoneum* (note that it's not the parietal abdominopelvic that just sounds weird).

The other parts of the body are divided into *axial* and *appendicular* areas. The axial portions are the parts of your body that form your axis — the head, chest, and abdomen. The appendicular portions form your appendages — your arms and legs. For consistency when referencing them, there are proper terms for all of the body's areas. The terminology used in identifying many of the regions is found in [Table 1-1](#). You'll notice these terms popping up all over this book.

Table 1-1 The Body's Regions

<i>Proper Term</i>	<i>Region</i>	<i>Proper Term</i>	<i>Region</i>
Antebrachial	forearm	Genicular	knee
Antecubital	inner elbow	Inguinal	groin/inner thigh
Axillary	armpit	Lumbar	lower back
Brachial	upper arm	Mental	chin
Buccal	cheek	Orbital	eye
Carpal	wrist	Otic	ear
Cephalic	head	Pectoral	chest
Cervical	neck	Pedal	foot
Coxal	hip	Plantar	sole/bottom of foot
Crural	shin	Popliteal	back of knee
Cubital	elbow	Sural	calf
Dorsum	back	Tarsal	ankle
Femoral	thigh	Vertebral	backbone
Frontal	forehead		

That's a lot of new terms for the first chapter! Let's see how well they're sticking.



EXAMPLE Q. Which of the following organs would you find in the mediastinum?

- I. lungs
 - II. heart
 - III. liver
- a. I only
 - b. II only
 - c. III only
 - d. I & II
 - e. I, II, & III

A. The correct answer is only the heart. The mediastinum is defined as the area between the lungs and the liver is in the abdominopelvic cavity.

1-10 Label the body cavities illustrated in [Figure 1-1](#).

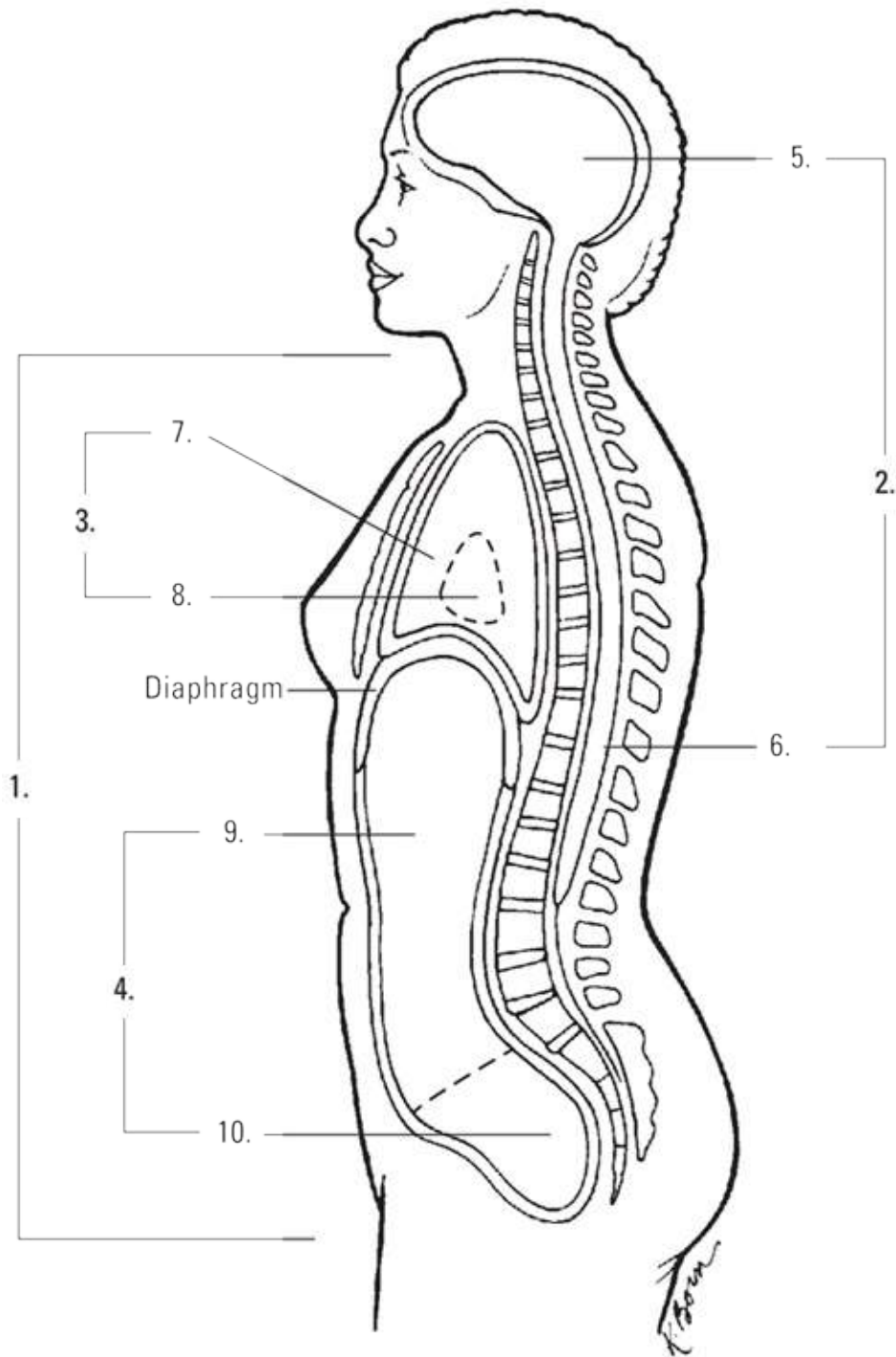


Illustration by Kathryn Born, MA

FIGURE 1-1: Body cavities.

- a. Abdominal
- b. Abdominopelvic
- c. Cranial

- d. Dorsal
- e. Pelvic
- f. Pericardial
- g. Pleural
- h. Spinal
- i. Thoracic
- j. Ventral

11-16 Match the description to identify the membranes that create the body's cavities.

- a. parietal pericardium
- b. parietal peritoneum
- c. parietal pleura
- d. visceral pericardium
- e. visceral peritoneum
- f. visceral pleura

- 11. _____ The outermost layer encasing the heart
- 12. _____ The membrane that lies on the surface of the liver
- 13. _____ The surface of the heart
- 14. _____ The lining of the thoracic cavity
- 15. _____ The membrane making direct contact with the lungs
- 16. _____ The layer that lines the abdominopelvic cavity

17 True or False: The cephalic region is considered part of the appendicular body.

18 Which body part would be affected if you injured your tarsal region?

- a. knee
- b. wrist

- c. ankle
- d. shoulder
- e. hip

19 If you suffered a laceration (cut) to your chin, the injury would be located in the ____ region.

- a. cubital
- b. cervical
- c. buccal
- d. mental
- e. frontal

20 Identify the correct pairing of terms:

- a. popliteal – inner elbow
- b. lumbar – back of the neck
- c. antecubital – upper arm
- d. coxal - shoulder
- e. sural – back of lower leg

Getting into Position

In anatomy and physiology, we often identify the body's features in reference to other body parts. Because of this, we need a standardized point of reference, which is known as *anatomical position*.



REMEMBER Anatomical position is the body facing forward, feet pointed straight ahead, arms resting on the sides, with the palms turned outward. Unless you are told otherwise, this is the body's position whenever specific body parts are described in reference to other locations.

Because we can only see the external surface of the body, sections must be

made in order for us to see what's inside. It's important to take note of what type of section was made to provide the view you see in a picture or diagram. There are three planes (directions) in which sections can be made:

- » **frontal:** separating the front from the back
- » **sagittal:** dividing right and left sides
- » **transverse:** creating top and bottom pieces

We also use directional terms to describe the location of structures. It helps to learn them as their opposing pairs to minimize confusion. The most commonly used terms are:

- » **anterior/posterior:** in front of/behind
- » **superior/inferior:** above/below
- » **medial/lateral:** closer to/further from the midline (also used with rotation)
- » **superficial/deep:** closer to/further from the body surface
- » **proximal/distal:** closer to/further from attachment point (used for appendages)



WARNING Right and left are also used quite often but be careful! They refer to the patient's right and left, not yours.

You got it? Let's find out.

21-23 Identify the planes of body sections in [Figure 1-2](#).

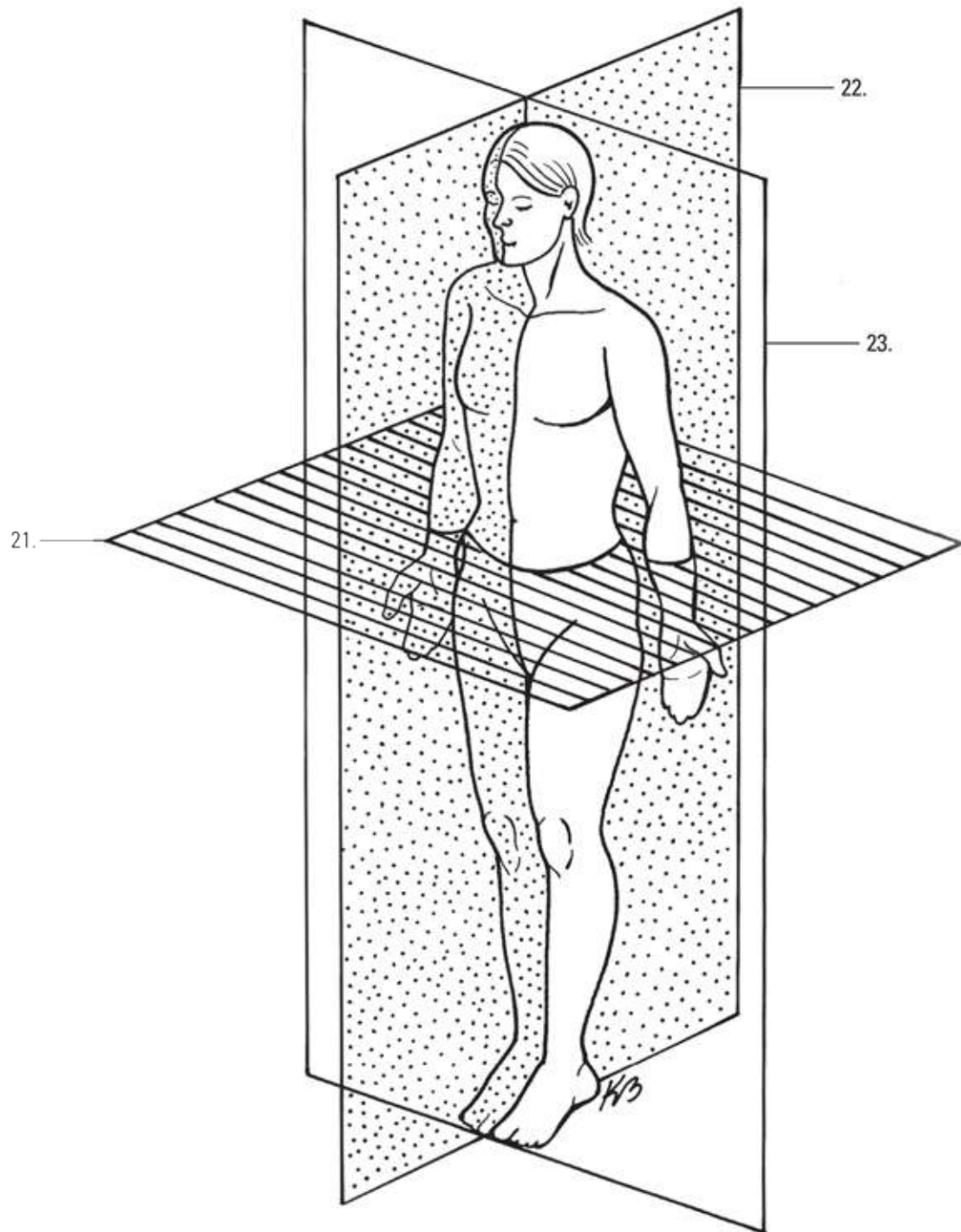


Illustration by Kathryn Born, MA

FIGURE 1-2: The body's planes.

- a. Sagittal
- b. Transverse
- c. Frontal

24-28 Fill in the blanks.

- 24. The neck is _____ to the hips.
- 25. The lungs are _____ to the rib cage.
- 26. The nose is _____ to the ears.
- 27. The wrist is _____ to the shoulder.
- 28. The buttocks are _____ to the navel (belly button).

Answers to Questions on Terminology

The following are answers to the practice questions presented in this chapter.

- 1-10 [Figure 1-1](#) should be labeled as follows:
 - 1. **j.** ventral, 2. **d.** dorsal, 3. **i.** thoracic, 4. **b.** abdominopelvic, 5. **c.** cranial, 6. **h.** spinal, 7. **g.** pleural, 8. **f.** pericardial, 9. **a.** abdominal, 10. **e.** pelvic
- 11 The outermost layer encasing the heart: **a. parietal pericardium**
- 12 The membrane that lies on the surface of the liver: **e. visceral peritoneum**
- 13 The surface of the heart: **d. visceral pericardium**
- 14 The lining of the thoracic cavity: **c. parietal pleura**
- 15 The membrane making direct contact with the lungs: **f. visceral pleura**
- 16 The layer that lines the abdominopelvic cavity: **b. parietal peritoneum**



TIP

Don't memorize all nine terms (cavities included), memorize the naming system. The space is always the cavity and the visceral layer is always making direct contact with an organ. The pattern holds true everywhere (except for surrounding the brain and spinal cord; they're special).

- 17 The cephalic region is considered part of the appendicular body. **False.**

The cephalic region is the head and though it does stick off the trunk, it's axial. Only the arms and legs are appendicular.

- 18) Which body part would be affected if you injured your tarsal region? **c. ankle**
- 19) If you suffered a laceration (cut) to your chin, the injury would be located in the **d. mental** region.
- 20) Identify the correct pairing of terms: **e. sural – back of lower leg**
- 21-23) [Figure 1-2](#) should be labeled as follows: **21. b. transverse, 22. a. sagittal, 23. c. frontal**
- 24) The neck is **superior** to the hips.
- 25) The lungs are **deep** to the ribcage.
- 26) The nose is **medial** to the ears.
- 27) The wrist is **distal** to the shoulder.
- 28) The buttocks are **posterior** to the navel (belly button).

Chapter 2

The Chemistry of Life

IN THIS CHAPTER

- » Getting to the heart of all matter: Atoms
 - » Checking into chemical reactions and compounds
 - » Making sense of metabolism
-

We can hear your cries of alarm. You thought you were getting ready to learn about the knee bone connecting to the thigh bone. How in the heck does that involve (horrors!) *chemistry*? As much as you may not want to admit it, chemistry — particularly *organic chemistry*, the branch of the field that focuses on carbon-based molecules — is a crucial starting point for understanding how the human body works. When all is said and done, the universe boils down to two fundamental components: *matter*, which occupies space and has mass; and *energy*, the ability to do work or create change. In this chapter, we review the interactions between matter and energy to give you some insight into what you need to know to ace those early-term tests.

Building from Scratch: Atoms and Elements

All matter — be it solid, liquid, or gas — is composed of atoms. An *atom* is the smallest unit of matter capable of retaining the identity of an element during a chemical reaction. An *element* is a substance that can't be broken down into simpler substances by normal chemical reactions (the ones on the periodic table that you may have had to memorize at some point). There are 98 naturally occurring elements in nature (though 10 of these have only ever been observed in trace amounts) and 20 (at last count) artificially created elements for a total of 118 known elements. The periodic table of elements organizes all the elements by name, symbol, atomic weight, and atomic number. The *bulk elements* of interest to students of anatomy and physiology

are

- » **Oxygen:** Symbol O
- » **Carbon:** Symbol C
- » **Hydrogen:** Symbol H
- » **Nitrogen:** Symbol N
- » **Phosphorus:** Symbol P
- » **Sulfur:** Symbol S



TIP

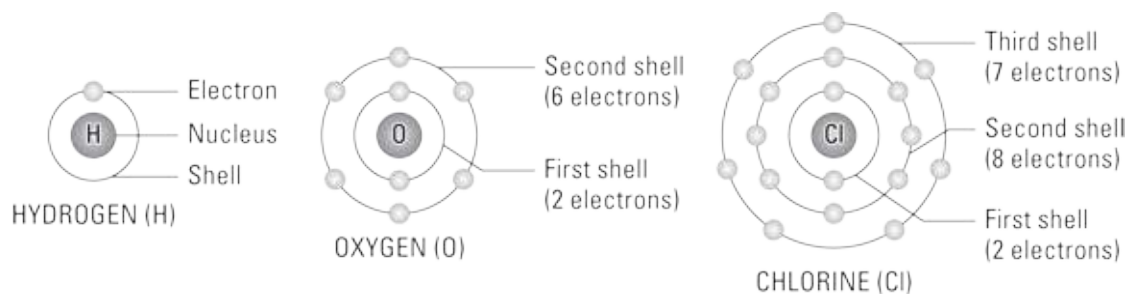
These six elements make up 95 percent of all living material. Just remember CHNOPS (read: chin-ops).



REMEMBER

Atoms are made up of the subatomic particles *protons* and *neutrons*, which are in the atom's *nucleus*, and clouds of *electrons* orbiting the nucleus. The *atomic weight*, or *mass*, of an atom is the total number of protons and neutrons in its nucleus. The *atomic number* of an atom is its number of protons; conveniently, atoms that are electrically neutral have the same number of positive charges as negative charges. Opposite charges attract, so negatively charged electrons are attracted to positively charged protons. The attraction holds electrons in orbits outside the nucleus. The more protons there are in the nucleus, the stronger the atom's positive charge is and the more electrons it can attract.

- » The first shell holds only two electrons.
- » The second and third shells hold eight electrons each.
- » The fourth shell (which can be found in elements such as potassium, calcium, and iron) holds up to 18 electrons. Higher shells also exist that hold even more electrons.



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FIGURE 2-1: Grouping electrons into shells or orbits.

Other key chemistry terms that you need to know are

- » **Isotopes:** Atoms of an element that have a different number of neutrons and a different atomic weight than usual. In other words, isotopes are alternate forms of the same chemical element, so they always have the same number of protons as that element but a different number of neutrons. The two most common in the body are potassium 40 and carbon 14.
- » **Ions:** Because electrons are relatively far from the atomic nucleus, they are most susceptible to external fields. Atoms that have gained or lost electrons are transformed into ions. Getting an extra electron turns an atom into a negatively charged ion, or *anion*, whereas losing an electron creates a positively charged ion, or *cation*. The four most prevalent are sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), and chloride (Cl^-).



TIP

To keep anions and cations straight, think like a compulsive dieter: Gaining is negative, and losing is positive.

- » **Acid:** A substance that becomes ionized when placed in solution, producing positively charged hydrogen ions, H^+ . An acid is considered a proton donor. (Remember, atoms always have the same number of electrons as protons. Ions are produced when an atom gains or loses electrons.) Stronger acids separate into larger numbers of H^+ ions in solution.
- » **Base:** A substance that becomes ionized when placed in solution, producing negatively charged hydroxide ions, OH^- . Bases are referred to as being more *alkaline* than acids and are known as proton acceptors.

Stronger bases separate into larger numbers of OH^- ions in solution.

» **pH (potential of hydrogen):** A mathematical measure on a scale of 0 to 14 of the acidity or alkalinity of a substance. A solution is considered *neutral*, neither acid nor base, if its pH is exactly 7. (Pure water has a pH of 7.) A substance is *basic* if its pH is greater than 7 and *acidic* if its pH is less than 7. The strength of an acid or base is quantified by its absolute difference from that neutral number of 7. This number is large for a strong base and small for a weak base. Interestingly, skin is considered acidic because it has a pH around 5. Blood, on the other hand, is slightly basic with a pH around 7.4.

Answer these practice questions about atoms and elements:

- 1 Which of the following is NOT a bulk element found in most living matter?
 - a. Carbon
 - b. Oxygen
 - c. Nitrogen
 - d. Potassium
 - e. Sulfur
- 2 Among the subatomic particles in an atom, the two that have equal weight are
 - a. neutrons and electrons.
 - b. protons and neutrons.
 - c. positrons and protons.
 - d. neutrons and positrons.
- 3 For an atom with an atomic number of 19 and an atomic weight of 39, the total number of neutrons is
 - a. 8.
 - b. 19.
 - c. 20.

d. 39.

e. 58.

4 Element X has 14 electrons. How many electrons are in its outermost shell?

a. 2

b. 4

c. 6

d. 8

e. 14

5 A substance that, in water, separates into a large number of hydroxide ions is

a. a weak acid.

b. a weak base.

c. a strong acid.

d. a strong base.

e. neutral.

6-11 Fill in the blanks to complete the following sentences:

Different isotopes of the same element have the same number of

6._____ and **7.**_____ but different numbers of

8._____. Isotopes also have different atomic **9.**_____.

An atom that gains or loses an electron is called a(n) **10.**_____.

If an atom loses an electron, it carries a(n) **11.**_____ charge.

Chemical Reactions

In the following sections, we describe the chemical bonds that hold together molecules and the organic compounds created by *chemical reactions*.

Chemical reactions are at the root of many of our physiological processes. A chemical reaction rearranges the atoms of the *reactant* molecules to generate new, *product* molecules. The three most common chemical reactions in our

bodies are:



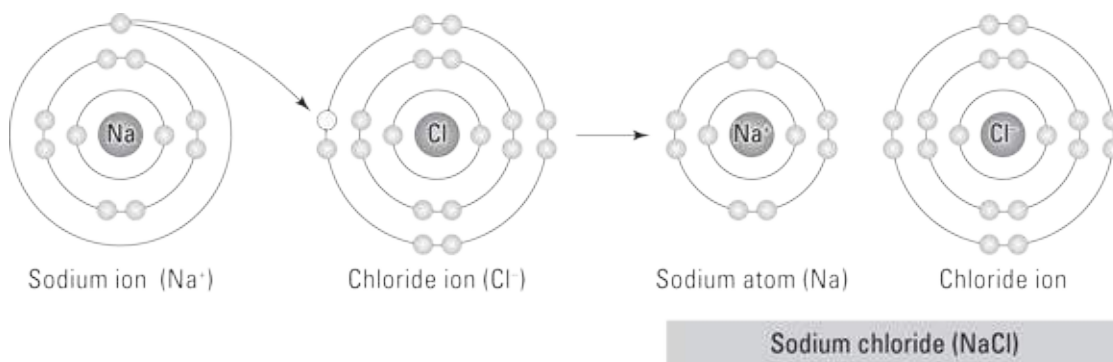
- » REMEMBER **Synthesis:** $A + B \rightarrow AB$; for example, the formation of proteins
- » **Decomposition:** $AB \rightarrow A + B$; for example, nutrient digestion
- » **Exchange:** $AB + CD \rightarrow AC + BD$; for example, buffers counteracting a pH change

Chemical bonds

Atoms tend to arrange themselves in the most stable patterns possible, which means that they have a tendency to complete or fill their outermost electron orbits. They join with other atoms to do just that. The force that holds atoms together in collections known as *molecules* is referred to as a *chemical bond*. Atoms of different elements have varying affinities for electrons, measured by their *electronegativity*. This dictates the type of chemical bonds they will form, which are described in the next few sections.

Ionic bond

This chemical bond (shown in [Figure 2-2](#)) involves a transfer of an electron to the atom with higher electronegativity. So one atom gains an electron while one atom loses an electron. One of the resulting ions carries a negative charge (anion), and the other ion carries a positive charge (cation). Because opposite charges attract, the atoms bond together to form a molecule.

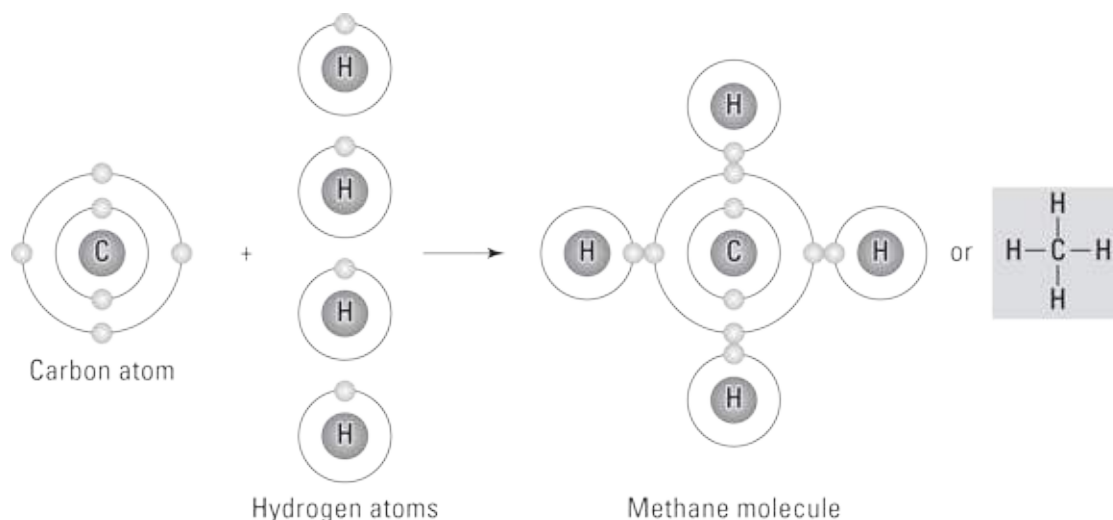


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FIGURE 2-2: Ionic bonding.

Covalent bond (non-polar)

The most common bond in organic molecules, a covalent bond (shown in [Figure 2-3](#)) involves the sharing of electrons between two atoms with similar electronegativities. In the human body, this is the most stable connection between atoms (though this does not hold true outside of the presence of water).



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FIGURE 2-3: Covalent bonding.

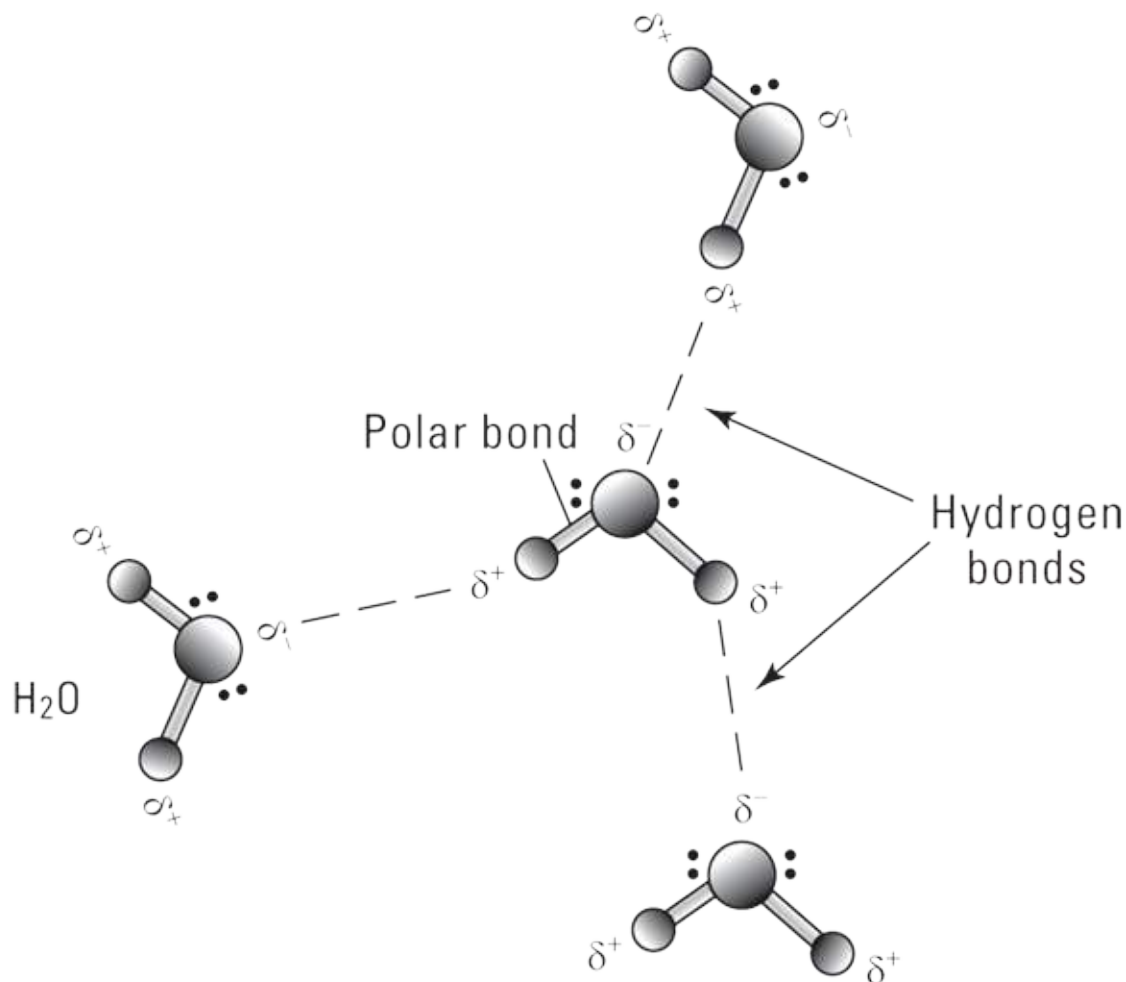
Polar covalent bond

Sometimes atoms with differing electronegativity will form a covalent bond, producing an unevenly distributed charge. This is known as a *polar bond* (shown in [Figure 2-4](#)), an intermediate case between ionic and covalent bonding, with one end of the molecule slightly negatively charged and the other end slightly positively charged. These slight imbalances in charge distribution are indicated in [Figure 2-4](#) by lowercase delta symbols (δ) with a charge superscript (+ or -). Although the resulting molecule is neutral, at close distances the uneven charge distribution can be important. Water (H₂O) is an example of a polar molecule; the oxygen end has a slight negative charge whereas the hydrogen ends are slightly positive. Polarity explains why some substances dissolve readily in water and others do not.

Hydrogen bond

Because they're polarized, two adjacent H₂O molecules (or any other polar molecules) can form a linkage known as a *hydrogen bond* (see [Figure 2-4](#)), where the (electropositive) hydrogen atom of one H₂O molecule is

electrostatically attracted to the (electronegative) oxygen atom of an adjacent water molecule. Consequently, molecules of water join together transiently in a hydrogen-bonded lattice. Hydrogen bonds have only about 1/20 the strength of a covalent bond, yet even this force is sufficient to affect the structure of water, producing many of its unique properties, such as high surface tension and specific heat. Hydrogen bonds are important in many life processes, such as in replication and defining the shape of DNA molecules.



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FIGURE 2-4: Polar and hydrogen bonds.

Organic compounds



REMEMBER

When different elements combine through chemical reactions, they

form *compounds*. When compounds contain carbon, they're called *organic compounds*. The four families of organic compounds with important biological functions are covered in the following sections.

Carbohydrates

These molecules consist of carbon, hydrogen, and oxygen in a ratio of roughly 1:2:1.

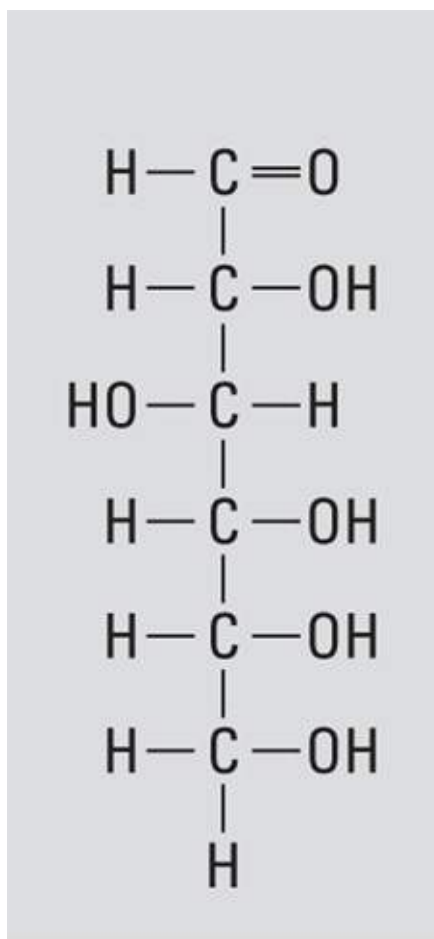


TIP

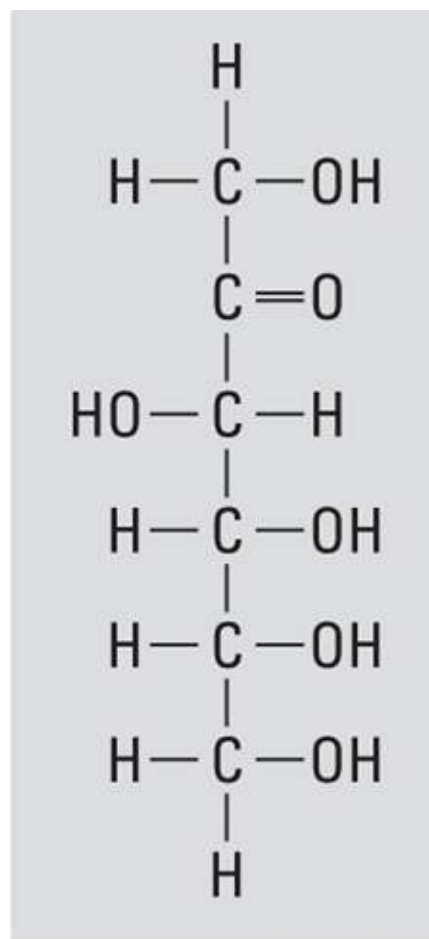
If a test question involves identifying a compound as a carbohydrate, count the atoms and see if they fit that ratio.

Carbohydrates are formed by the chemical reaction process of *condensation*, or *dehydration synthesis*, and broken apart by *hydrolysis*, the cleavage of a compound by a reaction that adds water. There are several subcategories of carbohydrates:

- » *Monosaccharides*, or *simple sugars*, are the building blocks, or *monomers*, of larger carbohydrate molecules and are a source of stored energy (see [Figure 2-5](#)). Key monomers include *glucose*, *fructose*, and *galactose*. These three have the same numbers of carbon (6), hydrogen (12), and oxygen (6) atoms in each molecule — formally written as $C_6H_{12}O_6$ — but the bonding arrangements are different. Molecules with this kind of relationship are called *isomers*. Two important five-carbon monosaccharides (pentoses) are *ribose*, a component of ribonucleic acids (RNA), and *deoxyribose*, a component of deoxyribonucleic acids (DNA).
- » *Disaccharides* are sugars formed by the bonding of two monosaccharides, including *sucrose* (table sugar), *lactose*, and *maltose*.



Glucose
(C₆H₁₂O₆)



Fructose
(C₆H₁₂O₆)

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FIGURE 2-5: Monosaccharides.

- » *Oligosaccharides* (from the Greek *oligo*, a few, and *sacchar*, sugar) contain three to nine simple sugars that serve many functions. They are found on plasma membranes of cells where they function in cell-to-cell recognition.
- » *Polysaccharides* are *polymers*, formed when many monomers bond into long, chainlike molecules. *Glycogen* is the primary polymer in the body; it breaks down into individual monomers of glucose, which cells use to generate usable energy.

Lipids

(a) Saturated Fatty Acids



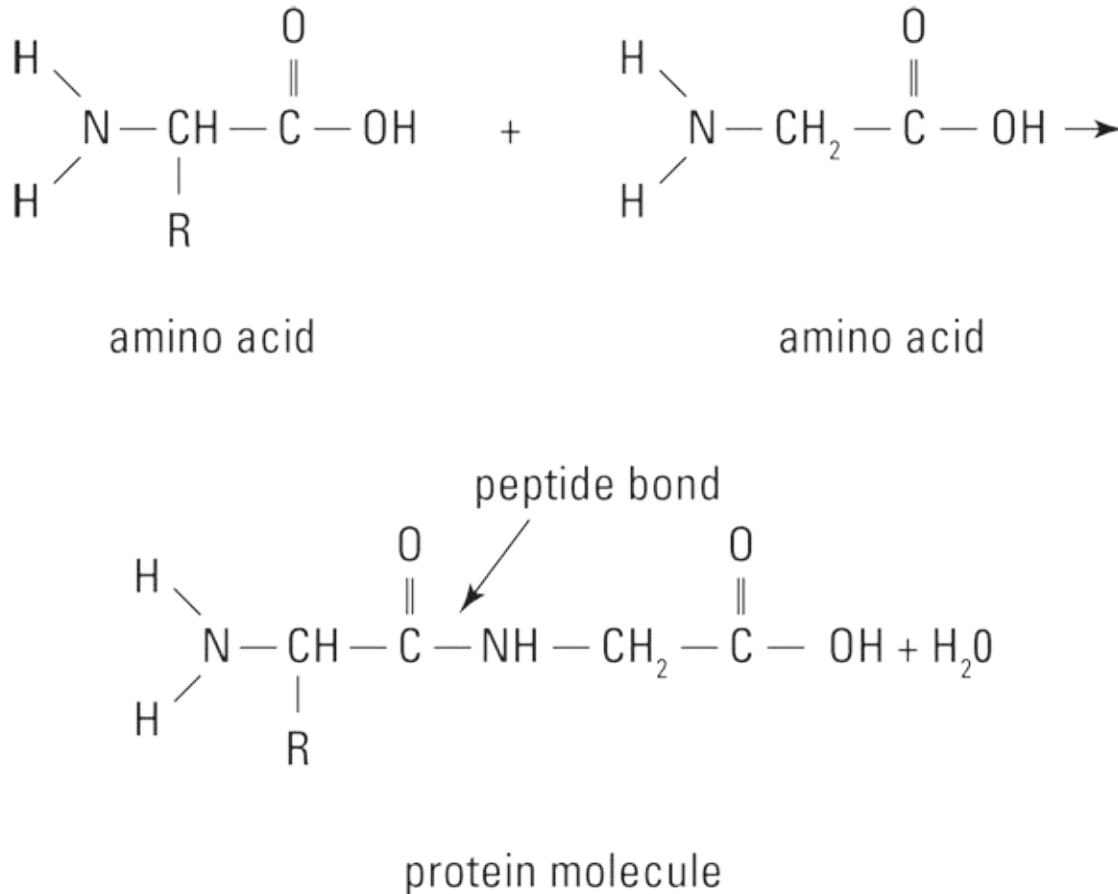
FIGURE 2-6: Fatty acids.

Proteins

Among the largest molecules, proteins can reach molecular weights of some 40 million atomic units. Proteins always contain hydrogen, oxygen, nitrogen, and carbon and sometimes contain phosphorus and sulfur. Examples of proteins in the body include *antibodies*, *hemoglobin* (the red pigment in red blood cells), and *enzymes* (catalysts that accelerate reactions in the body).

The human body builds protein molecules using 20 different kinds of monomers called *amino acids* (see [Figure 2-7](#)). An amino acid is a carbon atom attached to a hydrogen atom, an *amino group* ($-\text{NH}_2$), a *carboxyl group* ($-\text{COOH}$), and a unique side chain called the *R group*. Amino acids link together by *peptide bonds* to form long molecules called *polypeptides*, which then assemble into proteins. These bonds form when the carboxyl group of one molecule reacts with the amino group of another molecule, releasing a molecule of water (*dehydration synthesis*). A polypeptide, however, is not a

functioning protein. It must then be folded, twisted, and often linked with other polypeptides to create a three-dimensional structure which allows it to carry out its function.



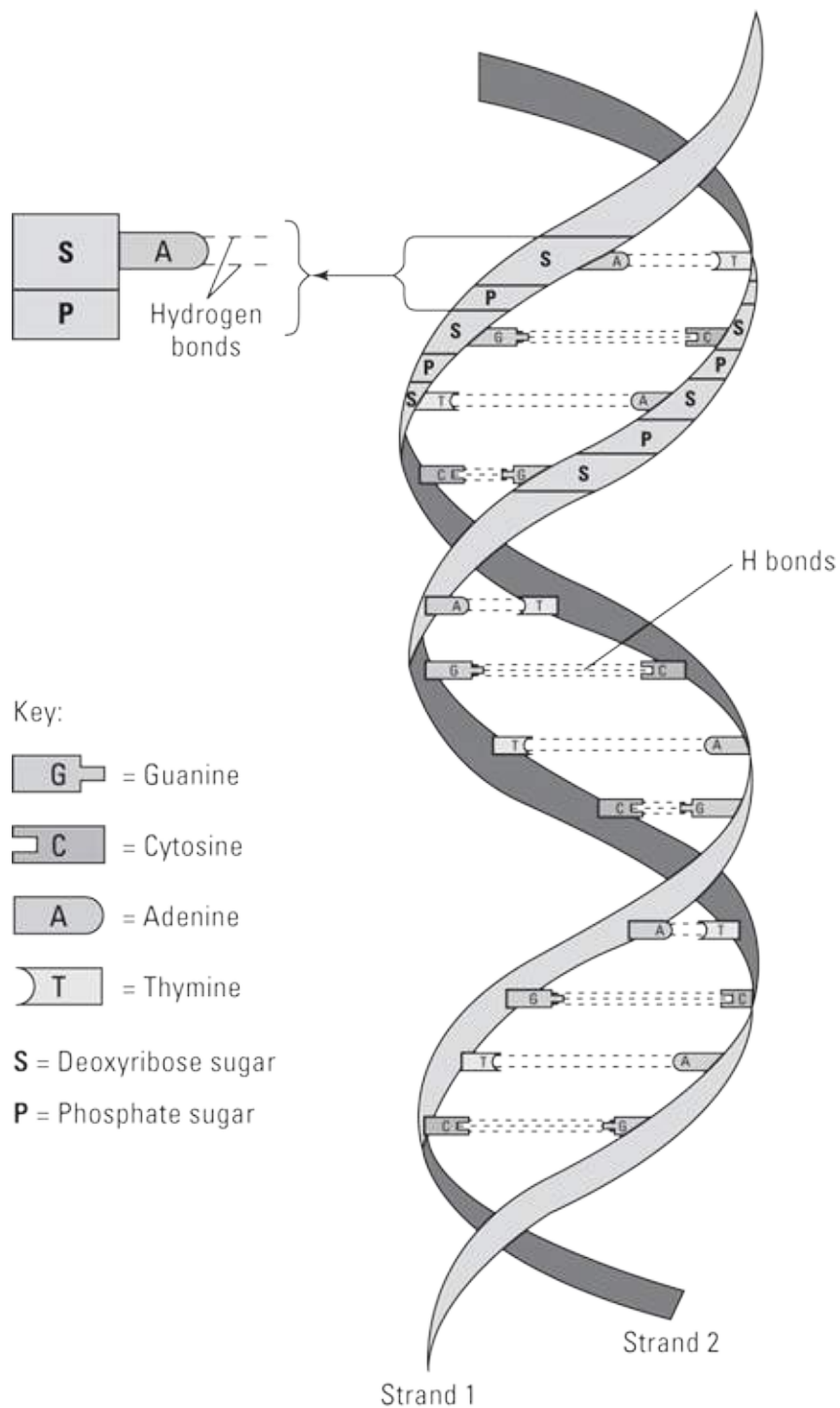
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FIGURE 2-7: Amino acids in a protein molecule.

Nucleic acids

These long molecules, found primarily in the cell's nucleus, act as the body's genetic blueprint. They're comprised of smaller building blocks called *nucleotides*. Each nucleotide, in turn, is composed of a five-carbon sugar (*deoxyribose* or *ribose*), a phosphate group, and a nitrogenous base. The sugar and phosphate groups link to form the backbone of the molecule. The base is attached to the sugar and aligns with its partner on the other strand (see [Figure 2-8](#)). The nitrogenous bases in DNA (deoxyribonucleic acid) are *adenine*, *thymine*, *cytosine*, and *guanine*; they always pair off A-T and C-G forming hydrogen bonds between the bases, creating the rungs of the DNA ladder. In RNA (ribonucleic acid), which occurs in a single strand, thymine is

replaced by *uracil*, so the nucleotides pair off A-U and C-G during transcription (see [Chapter 3](#) for more on this).



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FIGURE 2-8: The DNA double helix.

The following is an example question dealing with chemical reactions:



EXAMPLE

Q. Oxygen can react with other atoms because it has

- a. two electrons in its inner orbit.
- b. eight protons.
- c. an incomplete outer electron orbit.
- d. eight neutrons.

A. The correct answer is an incomplete outer electron orbit. Even if you don't know the first thing about oxygen, remembering that atoms tend toward stability answers this question for you.

12 Covalent bonds are a result of

- a. completing an inner electron orbit.
- b. sharing one or more electrons between atoms.
- c. deleting neutrons from both atoms.
- d. reactions between protons.
- e. transferring electrons to another atom.

13 The formation of chemical bonds is based on the tendency of an atom to

- a. move protons into vacant electron orbit spaces.
- b. fill its outermost energy level.
- c. radiate excess neutrons.
- d. pick up free protons.

14-19 Fill in the blanks to complete the following:

Molecules like water are **14.**_____ because they share their electrons unequally. This is due to oxygen having a higher **15.**_____ than hydrogen. As a result, the oxygen side of the molecule has a **16.**_____ charge while the hydrogen sides have a **17.**_____

charge. When exposed to other polar molecules **18.**_____ charges attract and they form **19.**_____ bonds.

20 Which of the following statements is *not* true of DNA?

- a. DNA is found in the nucleus of the cell.
- b. DNA can be replicated.
- c. DNA contains the nitrogenous bases adenine, thymine, guanine, cytosine, and uracil.
- d. DNA forms a double-helix molecule.
- e. DNA is made of monomers called nucleotides.

21 Polysaccharides

- a. can be reduced to fatty acids.
- b. contain nitrogen and phosphorus.
- c. are complex carbohydrates.
- d. are monomers of glucose.
- e. contain adenine and uracil.

22 Amino acids

- a. help reduce carbohydrates.
- b. are the building blocks of proteins.
- c. modulate the production of lipids.
- d. catalyze chemical reactions.
- e. control nucleic acids.

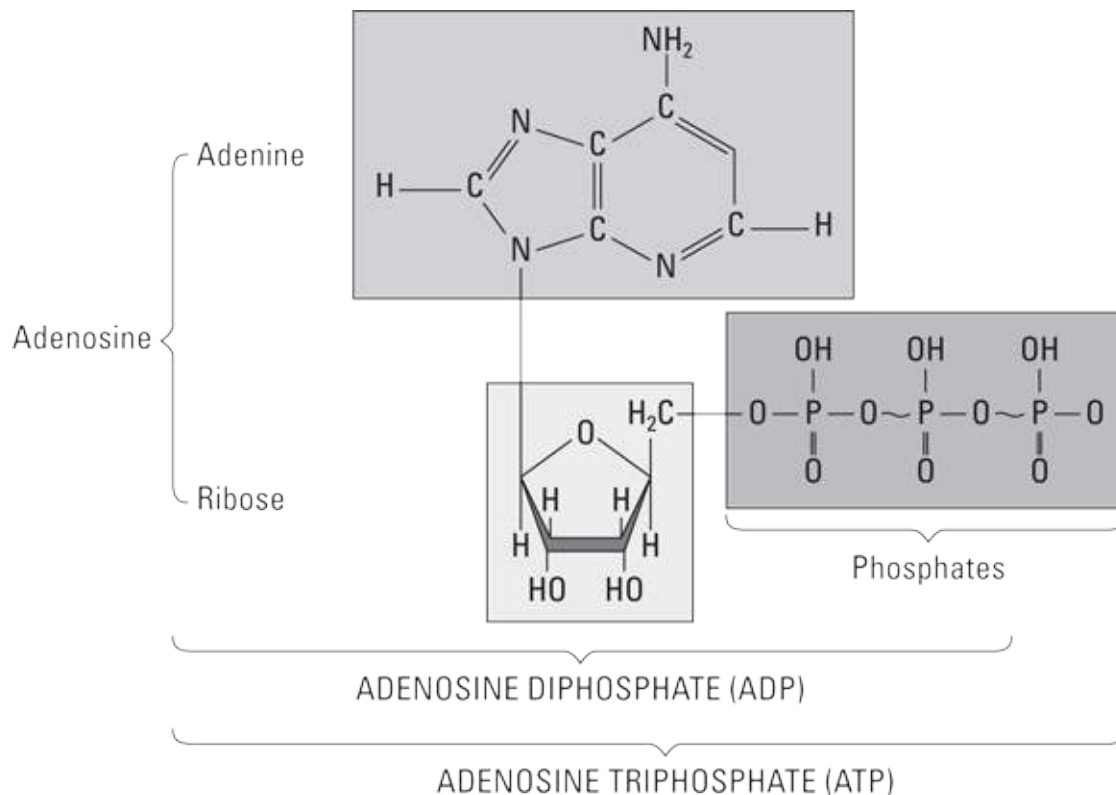
Cycling through Life: Metabolism

Metabolism (from the Greek *metabole*, which means “change”) is the term for all the chemical reactions our body must undergo to sustain life. Proteins called *enzymes* control the rate of these reactions that are of two types:

- » **Anabolic reactions** require a source of energy to build up compounds that the body needs.
- » **Catabolic reactions** break down larger molecules into smaller ones, releasing energy (Memory tip: It can be *catastrophic* when things break down).

The chemical alteration of molecules in the cell is referred to as *cellular metabolism*. Enzymes are biological catalysts that accelerate chemical reactions without being changed. Thus, one way we can control the rate of our cellular metabolism is through the production of enzymes (or lack of production). In this section, we discuss the processes that take energy from the food we eat and turn it into energy that a cell can actually use. To be fair, there are numerous other metabolic reactions occurring as well — like the breakdown of wastes in the liver and the building of structural proteins — but those are discussed in their relevant chapters.

Before we dive into the energy from food, we first need to discuss the energy that cells can use — a nucleic acid called *adenosine triphosphate* (ATP). As the *tri-* prefix implies, a single molecule of ATP is composed of three phosphate groups attached to adenosine (a nitrogenous base of adenine paired with ribose). ATP's energy is stored in the bonds that attach the second and third phosphate groups. (The high-energy bond is symbolized by a wavy line in [Figure 2-9](#).) When a cell needs energy, it removes one or two of these phosphate groups, releasing energy and converting ATP into either the two-phosphate molecule *adenosine diphosphate* (ADP) or the one-phosphate molecule *adenosine monophosphate* (AMP). Later, through additional metabolic reactions, the second and third phosphate groups are reattached to adenosine, reforming an ATP molecule until energy is needed again. Thus, ATP works like a rechargeable battery. (**Note:** Most often, we cycle through ADP and ATP; AMP is only utilized in special scenarios.)



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FIGURE 2-9: The structures of ADP and ATP.

Energy from food

ATP is created via *cellular respiration*, a process that occurs in and around the mitochondria of a cell (you can see this entire process outlined in [Figure 2-10](#)). This is a series of reactions that includes glycolysis, the Krebs cycle, and the electron transport chain. It starts with a single molecule of glucose and produces 38 molecules of ATP. This process, though, requires energy sources of its own, which we gain from the foods we eat. The most efficient source (glucose) is gained from *carbohydrate metabolism*. When that is used up, we turn to the products of *lipid metabolism* and finally to the products of *protein metabolism*. Each energy source enters the cellular respiration process at different points (shown in [Figure 2-10](#)).

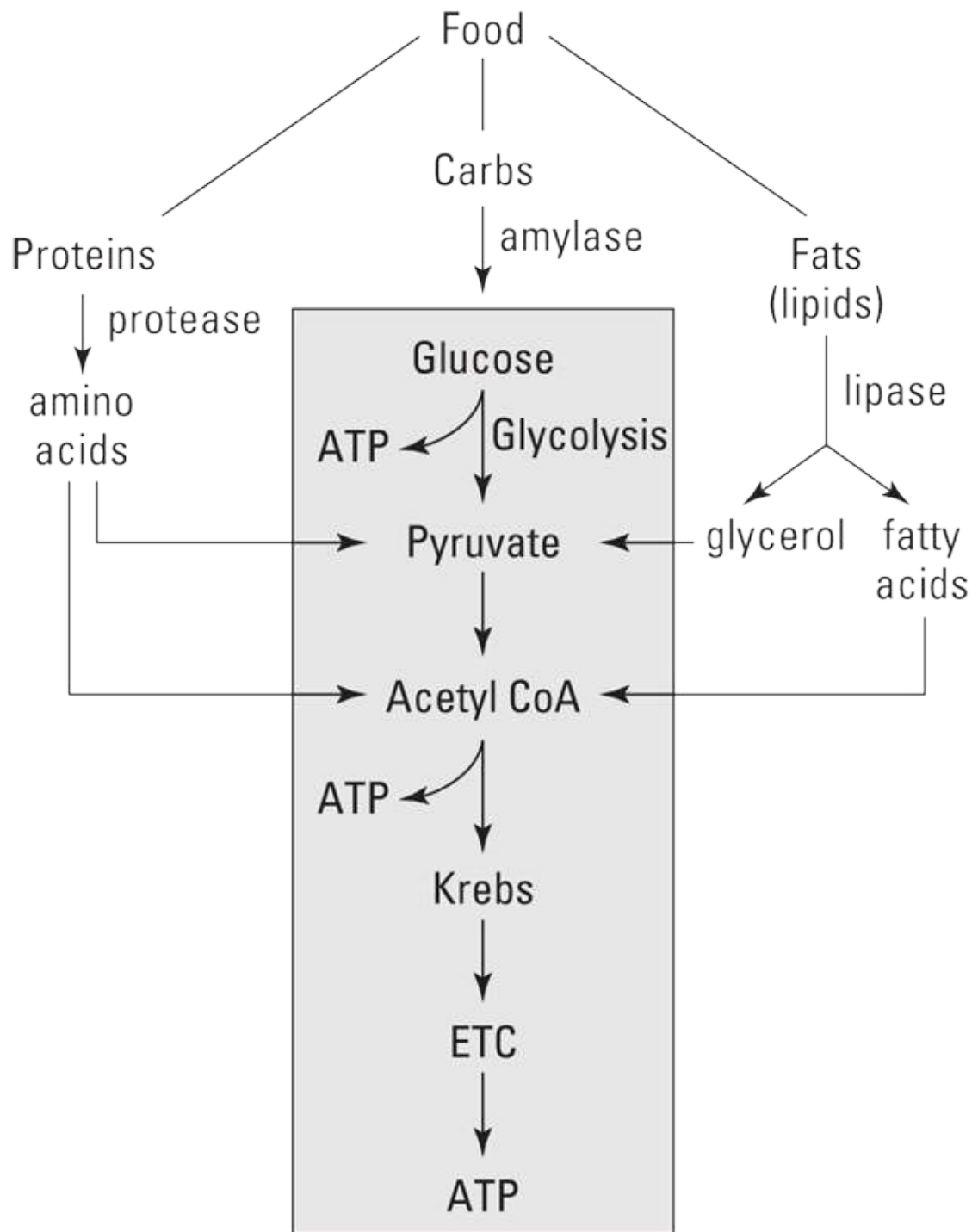
Carbohydrate metabolism

The metabolism of all carbohydrates leads to the eventual production of glucose. The glucose is then utilized in the first phase of cellular respiration, *glycolysis*.

From the Greek *glyco* (sugar) and *lysis* (breakdown), glycolysis is the process

that takes a glucose (six carbons) molecule and splits it into two molecules of *pyruvate* (three carbons each). The pyruvate, or *pyruvic acid*, will then enter the Krebs cycle to finish carrying out cellular respiration. This process, called phosphorylation, requires the energy from two molecules of ATP. However, by the end of the process we have a net gain of two ATP. In addition to the two pyruvates produced, it also yields two molecules of NADH (another energy molecule), both of which will be used later in cellular respiration.

Glycolysis takes place in the cell's cytoplasm and doesn't require oxygen to occur. The pyruvate and NADH move into the cell's mitochondria (detailed in [Chapter 3](#)), where an aerobic (with oxygen) process uses them to create ATP.



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FIGURE 2-10: Protein, carbohydrate, and fat metabolism.

Lipid metabolism

Lipids contain about 99 percent of the body's stored energy and can be digested at mealtime, but as people who complain about fats going "straight

to their hips” can attest, lipids are more inclined to be stored in *adipose tissue* — the stuff generally identified as body fat. When the body is ready to metabolize lipids, an enzyme called *lipase* splits them into their monomer units: glycerol and fatty acid chains. Glycerol, which has three carbons, is then converted into pyruvate. A different series of catabolic reactions breaks apart two carbon atoms from the end of a fatty acid chain to form acetyl CoA, which then enters the Krebs cycle to produce ATP. Those reactions continue to strip two carbon atoms at a time until the entire fatty acid chain is converted into acetyl CoA molecules.

Protein metabolism

Protein metabolism focuses on producing the amino acids needed for synthesis of protein molecules within the body. But, like the lipids, the amino acids can be converted into pyruvate and acetyl CoA. Additionally, protein metabolism generates energy molecules such as NADH and FADH₂ that power the final step of cellular respiration — the electron transport chain.

Getting energy from energy

Now we take a look at how we can utilize the metabolites of carbohydrate, lipid, and protein metabolism to generate ATP, the cell’s energy currency. This occurs through the final two steps of cellular respiration: the Krebs cycle and the electron transport chain (the first step, glycolysis, was discussed in the carbohydrate metabolism section). You can see the entire process detailed in [Figure 2-11](#).

Krebs cycle

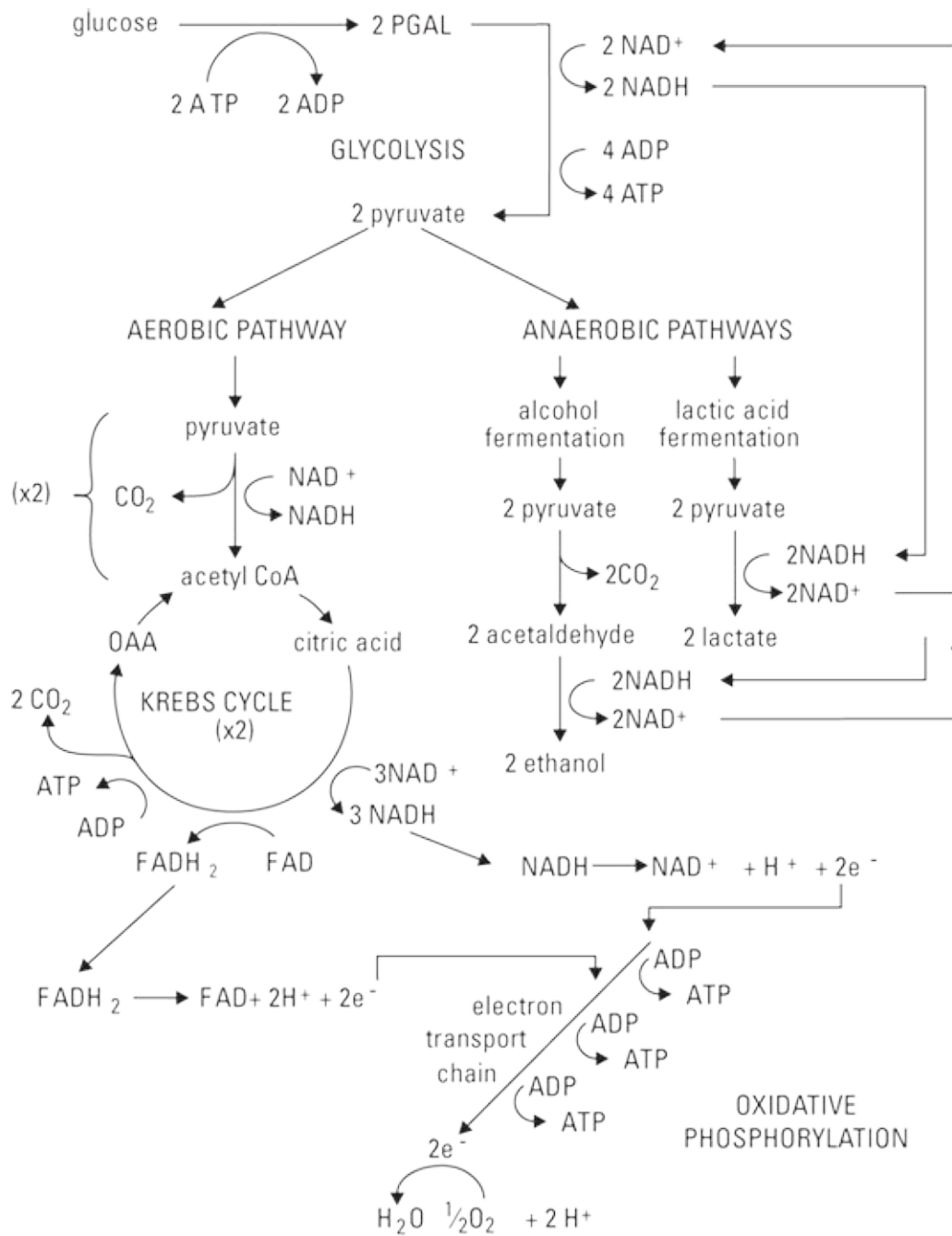
Also known as the *citric acid cycle*, this series of energy-producing chemical reactions begins in the mitochondria after pyruvate arrives. Before the Krebs cycle can begin, the pyruvate loses a carbon dioxide group to form *acetyl coenzyme A* (acetyl CoA). Then acetyl CoA combines with a four-carbon molecule (*oxaloacetic acid*, or OAA) to form a six-carbon citric acid molecule that then enters the Krebs cycle. The CoA is released intact to bind with another acetyl group, reforming the acetyl CoA to be used with the next pyruvate.

The cycle goes through eight steps, rearranging the atoms of citric acid to produce different intermediate molecules called *keto acids*. This generates one ATP (for each pyruvate) and four energy molecules (three NADH and

one FADH_2) which power the electron transport chain. In the final step of the Krebs cycle, OAA is regenerated to get the next cycle going, and the carbon dioxide produced during this cycle is released from the cell to be exhaled from the lungs.

Electron transport chain

The electron transport chain, which requires oxygen, is the final step of cellular respiration and generates the bulk of the ATP. The chain itself is a series of enzyme complexes aligned on the inner folds of the mitochondria. Electrons from the energy molecules NADH and FADH_2 are passed into the chain and bounced from one protein to the next much like a game of hot potato. At the end of the chain, a molecule of H_2O is created and the energy from the bouncing electrons power *oxidative phosphorylation* — where an enzyme called *ATP synthase* reattaches a phosphate group to ADP, creating ATP. For every glucose molecule metabolized, 34 molecules of ATP are generated by the electron transport chain.



Respiration

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FIGURE 2-11: Cellular respiration: Glycolysis, aerobic (Krebs cycle) and anaerobic respiration, and oxidative phosphorylation, all of which convert energy from fuel into ATP.

One last thing: That severe soreness and fatigue you feel in your muscles after strenuous exercise is the result of lactic acid buildup during *anaerobic*

respiration (shown in [Figure 2-11](#)). Glycolysis continues because it doesn't need oxygen to take place. But glycolysis does need a steady supply of NAD^+ , which usually comes from the oxygen-dependent electron transport chain. In its absence, the body begins a process called *lactic acid fermentation*, in which one molecule of pyruvate combines with one molecule of NADH to produce ATP, a molecule of NAD^+ , plus a molecule of the toxic byproduct lactic acid.

While useful for energy creation in muscles when O_2 has been used up, it hardly seems worth it — anaerobic respiration provides only two ATP molecules per glucose compared to the 38 of aerobic respiration. Tack on the creation of lactic acid, which causes *muscle fatigue* (inability of fibers to contract), as a byproduct and you may wonder why this process even exists! Well, we need to exercise, and some of the muscle fibers (but not all) will inevitably run out of oxygen. Lucky for us, the liver has a built-in mechanism to rid the body of lactic acid, though it may not happen as quickly as we'd like.

Hopefully you're not too fatigued to try out a few questions about metabolism:



EXAMPLE

Q. Cells obtain ATP by converting the energy in

- a. carbohydrates.
- b. proteins.
- c. lipids.
- d. all of these.

A. The correct answer is all of these. While it's true that carbohydrates provide the most immediately available energy, proteins and lipids also contribute to the production of ATP.

23 During glycolysis, a molecule of glucose is

- a. broken down to two pyruvates.
- b. broken down to two lipids.

- c. converted into energy.
- d. used to oxidize proteins.
- e. used to release oxygen.

24 By the end of cellular respiration, a single molecule of glucose can be converted into how many ATP molecules?

- a. 2
- b. 4
- c. 24
- d. 38
- e. 42

25 The buildup of lactic acid can lead to all of the following EXCEPT:

- a. achy feeling muscles.
- b. tired feeling muscles.
- c. muscle cramps.
- d. inability of fibers to contract.

26 Which two respiration processes take place in the cell's mitochondria?

- a. Glycolysis and the Krebs cycle
- b. Glycolysis and the electron transport chain
- c. The Krebs cycle and the electron transport chain
- d. The Krebs cycle and lactic acid fermentation
- e. Lactic acid fermentation and glycolysis

27 To provide needed reactants for cellular respiration, cells utilize metabolites of the organic molecules we consume in the following order:

- a. lipids, proteins, carbohydrates
- b. lipids, carbohydrates, proteins

- c. proteins, lipids, carbohydrates
- d. carbohydrates, proteins, lipids
- e. carbohydrates, lipids, proteins

28 What are the primary products of protein metabolism?

- a. ATP molecules
- b. Amino acids
- c. Lipids
- d. Simple sugars
- e. Carbon dioxide molecules

Answers to Questions on Life's Chemistry

The following are answers to the practice questions presented in this chapter.

- ① Which of the following is NOT a bulk element found in most living matter:
d. potassium
- ② Among the subatomic particles in an atom, the two that have equal weight are **b. protons and neutrons**. That's why you add them together to determine atomic weight, or mass.
- ③ For an atom with an atomic number of 19 and an atomic weight of 39, the total number of neutrons is **c. 20**. The atomic number of 19 is the same as the number of protons. The atomic weight of 39 tells you the number of protons plus the number of neutrons: $39 - 19 = 20$.
- ④ Element X has 14 electrons. How many electrons are in its outermost shell? **b. 4**. The first orbit has the maximum two electrons, and the second orbit has the maximum eight electrons. That makes ten electrons in the first two orbits, leaving only four for the third, outermost orbit.
- ⑤ A substance that, in water, separates into a large number of hydroxide ions

is **d. a strong base**. The more hydroxide ions there are, the stronger the base is.

- 6-11 Different isotopes of the same element have the same number of **6. electrons/protons** and **7. protons/electrons** but different numbers of **8. neutrons**. Isotopes also have different atomic **9. weights**. An atom that gains or loses an electron is called an **10. ion**. If an atom loses an electron, it carries a **11. positive** charge.
- 12 Covalent bonds are a result of **b. sharing one or more electrons between atoms**. If the atoms had gained or lost electrons, it would be an ionic bond, but here they're sharing — valiantly cohabiting, if you will.
- 13 The formation of chemical bonds is based on the tendency of an atom to **b. fill its outermost energy level**. This is true whether an atom fills its outer shell by sharing, gaining, or losing electrons.
- 14-19 Molecules like water are **14. polar** because they share their electrons unequally. This is due to oxygen having a higher **15. electronegativity** than hydrogen. As a result, the oxygen side of the molecule has a **16. partial negative** charge while the hydrogen sides have a **17. partial positive** charge. When exposed to other polar molecules **18. opposite** charges attract and they form **19. hydrogen** bonds.
- 20 Which of the following statements is *not* true of DNA? **c. DNA contains the nitrogenous bases adenine, thymine, guanine, cytosine, and uracil**. This statement is false because only RNA contains uracil.
- 21 Polysaccharides **c. are complex carbohydrates**. The root *poly*– means “many,” which you can interpret as “complex.” The root *mono*– means “one,” which you can interpret as “simple.”
- 22 Amino acids **b. are the building blocks of proteins**. Being such large molecules, proteins need to be built from complex molecules to begin with.
- 23 During glycolysis, a molecule of glucose is **a. broken down to two pyruvates**. Remember that glucose must become pyruvic acid before it enters the Krebs cycle.

- 24) By the end of cellular respiration, a single molecule of glucose can be converted into how many ATP molecules? **d. 38.** Two net molecules of ATP come from glycolysis, two molecules come from the Krebs cycle, and the electron transport chain churns out 34.
- 25) The buildup of lactic acid can lead to all of the following EXCEPT: **c. muscle cramps.** Cramps are caused by sustained fiber contraction. Lactic acid buildup does just the opposite: It prevents the fiber from contracting.
- 26) Which two respiration processes take place in the cell's mitochondria? **c. The Krebs cycle and the electron transport chain.** The other answers are incorrect because glycolysis takes place in the cytoplasm, and lactic acid fermentation isn't one of the three cellular respiration processes.
- 27) To provide needed reactants for cellular respiration, cells utilize metabolites of the organic molecules we consume in the following order: **e. carbohydrates, lipids, proteins.** This is why it can be so difficult to burn fat by exercising, as your body will first utilize any carbohydrates that are around before breaking down that stubborn fat.
- 28) The primary products of protein metabolism are **b. amino acids.** Although some ATP comes from metabolizing proteins, the body primarily needs to get amino acids from any protein that's consumed.

Chapter 3

The Cell: Life's Basic Building Block

IN THIS CHAPTER

- » Looking at the life of a cell
 - » Breaking through the cell membrane
 - » Moving toward the nucleus
 - » Sorting through what's inside the cell
 - » Putting together proteins made to order
-

Cytology, from the Greek word *cyto*, which means “cell,” is the study of cells. Every living thing has cells, but not all living things have the same kinds of cells. *Eukaryotes* like humans (and all other organisms besides bacteria and viruses) have *eukaryotic cells*, each of which has a defined *nucleus* that controls and directs the cell's activities, and *cytosol*, fluid material found in the gel-like *cytoplasm* that fills most of the cell. Plant cells have fibrous cell walls; animal cells have a semipermeable cell membrane called a *plasma membrane* or the *plasmalemma*. Because human cells don't have cell walls, they look like gel-filled sacs with nuclei and tiny parts called *organelles* nestled inside when viewed through an electron microscope.

In this chapter, we help you sort out what makes up a cell, what all those tiny parts do, and how cells act as protein-manufacturing plants to support life's activities.

Life and Death of a Cell

At any given time, an adult has between 50–100 trillion different cells. They come in about 260 different varieties, though they all originated from that single cell in which you began your life. Pretty amazing if you stop and think about it — especially because they all have the same DNA! Cells take on

their specialized structure and function through the process of *differentiation*. Most often, chemical triggers lead to differing *gene expression*; that is, some genes are turned on while others are turned off. This is how cardiac muscle cells and nervous cells can be so dramatically different despite having the same DNA. The field of *epigenetics* strives to understand how gene expression is controlled and there is still much to be learned!

Cell differentiation is often a process with numerous intermediaries between the *stem cell* and the final, specialized one. A stem cell is an undifferentiated cell that can give rise to cells that will then specialize (or they can just make more of themselves). Only very early in embryonic development do you find truly *totipotent* stem cells — those that can develop into any cell type found in our bodies, including that of the placenta. As the embryo begins to organize, the stem cells are now *pluripotent*, which means they can develop into any cell type except the placenta. Once the embryo starts to form its layers (the endoderm, mesoderm, and ectoderm), the stem cells are now considered *multipotent*. (For more on embryonic development, refer to [Chapter 17](#).) They have now started to follow one of three pathways: Cells in the mesoderm can develop into muscles and bones (among other things) but cannot become skin; cells in the ectoderm are on that path.

Differentiated cells then go on about their lives, carrying out their designated function until they either divide or die (though some cells like skeletal muscle and neurons share your life span so technically they do neither). Some cells, such as bone cells, for example, do not undergo mitosis. New ones must come from stem cells. Once a cell reaches its genetically determined expiration date, it begins the process of programmed cell death, or *apoptosis*. This continuous step-wise process ultimately leads to the cell shattering.

Let's see if you can sort out all of that terminology:

1-5 Match the terms with their descriptions.

- a. apoptosis
- b. differentiated
- c. multipotent
- d. pluripotent
- e. totipotent

1. _____ Cell that can become any type of other cell
2. _____ Partially differentiated stem cell; can still become numerous types but is limited
3. _____ Sequence of events that leads to the death of a cell
4. _____ Cell that carries out a specific specialized function
5. _____ Cell that can become most other cells, except placenta

Gaining Admission: The Cell Membrane

Think of it as a gatekeeper, guardian, or border guard. Despite being only 6 to 10 nanometers thick and visible only through an electron microscope, the cell membrane keeps the cell's cytoplasm in place and lets only select materials enter and depart the cell as needed. This *semipermeability*, or *selective permeability*, is a result of a double layer (bilayer) of *phospholipid* molecules interspersed with protein molecules. The outer surface of each layer is made up of tightly packed *hydrophilic* (or water-loving) *polar heads*. Inside, between the two layers, you find *hydrophobic* (or water-fearing) *nonpolar tails* consisting of fatty acid chains. (See [Figure 3-1](#).)

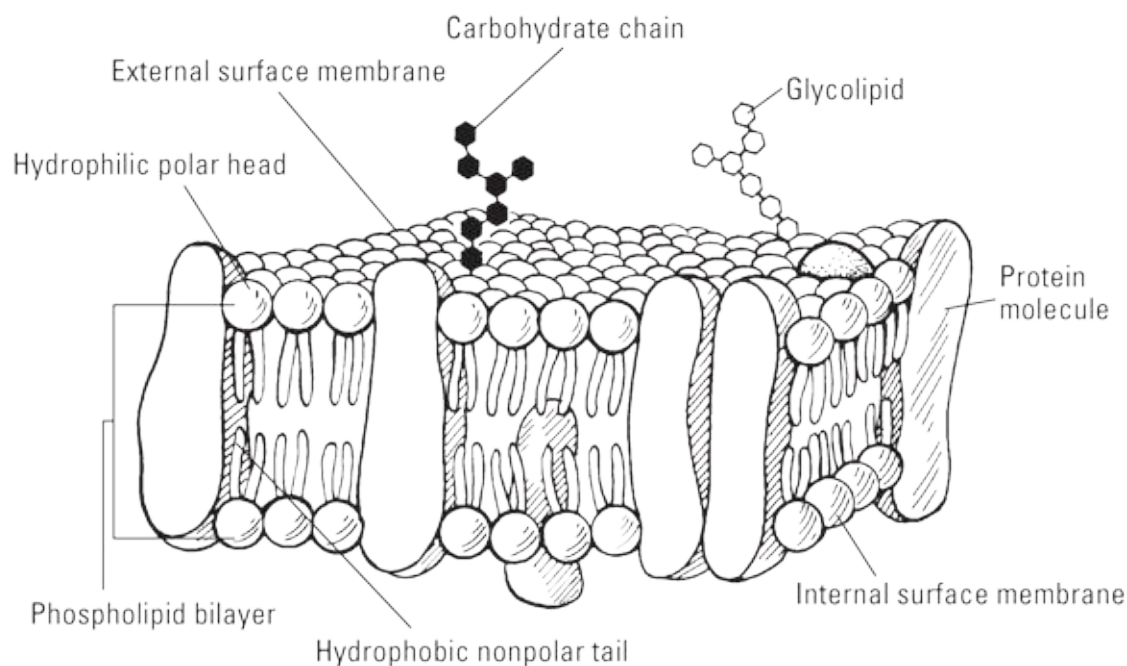


Illustration by Kathryn Born, MA

FIGURE 3-1: The cell membrane.

Since the phospholipids give the membrane a rather fluid structure, cholesterol molecules are embedded to provide stability and make it less permeable to water-soluble substances. Both cytoplasm and the *matrix*, the environment outside of the cell, are primarily water. The polar heads electrostatically attract polarized water molecules. The membrane's interior is made up of non-polar fatty acid molecules (the phospholipid tails) creating a dry middle layer. Lipid-soluble molecules can pass through this layer, but water-soluble molecules such as amino acids, sugars, and many proteins cannot. These molecules must enter the cell through little tunnels created by channel proteins. In addition to forming channels, other proteins do not span the membrane but stick out from the outer surface, sometimes with a carbohydrate attached. These serve numerous purposes: for example, creating receptors for hormones, binding sites for other cells, and aiding in recognition of self versus non-self.

The cell membrane is designed to hold the cell together and to isolate it as a distinct functional unit. Although it can spontaneously repair minor tears, severe damage to the membrane will cause the cell to disintegrate. The membrane is picky about which molecules it lets in or out. It allows movement across its barrier by *diffusion*, *osmosis*, or *active transport*.

Diffusion

Atoms, ions, and molecules are in constant motion, taking a straight-line path until they inevitably crash into another particle and bounce off with a new trajectory. As a result, particles will naturally flow to areas of lower concentration (more collisions means more movement). This is *diffusion* — the movement of molecules from high to low concentration. As long as the molecules can move through any barricade that may be in their way (the cell membrane, for instance), the body doesn't need to invest any energy in their movement. The kinetic energy of the particles is the driving force moving them down their *concentration gradient*. Even once *equilibrium* has been reached (molecules are equally distributed throughout), diffusion continues; there just isn't a net change (one randomly moves out so another moves in).

So what does this have to do with cells? Diffusion is an easy way to move things into and out of a cell, passing through the cell membrane. Because it requires no energy, it is a form of *passive transport*. Small, non-polar

molecules such as O₂ and CO₂ simply squeeze between phospholipid heads either entering or exiting the cell (depending on where the concentration of the molecule is lower). Since the inside of the membrane is hydrophobic, polar molecules and ions are unable to simply diffuse in (or out). They can, however, still move passively down their concentration gradient. They just require a protein channel to move through. This is termed *facilitated diffusion* as these molecules, like glucose, for example, need a protein to facilitate (allow for) their movement. Unlike the lipid-soluble molecules that use simple diffusion, we can control these water-soluble ones by opening and closing the protein channels.

Osmosis

Osmosis, simply put, is the diffusion of water. Since our bodies are 50–65 percent water, its movement is of the utmost importance to us. More specifically, osmosis is the passive diffusion of solvent through a selectively permeable membrane from an area of higher concentration to an area of lower concentration. Solutions are composed of two parts: a solvent and a solute.

- » The *solvent* is the liquid in which a substance is dissolved; in our bodies, this is always water.
- » A *solute* is the substance dissolved in the solvent.

Typically, a cell contains a roughly 1 percent saline solution — in other words, 1 percent salts (solutes) and 99 percent water (solvent). Water is a polar molecule that will not pass through the lipid bilayer; however, embedded in many cell membranes are numerous tiny proteins called *aquaporins*. Water readily moves through these little holes via osmosis, but many solutes do not (as they're too big). Because of these pores, we can't directly control the flow of water.

Transport by osmosis is influenced by the solute concentration of the water, the solution's *osmolarity*. In our bodies, water easily moves to area of higher osmolarity. Or in other words, water always goes to where there is less water. As water diffuses into a cell, *hydrostatic pressure* builds within the cell. Eventually, the pressure within the cell becomes equal to, and is balanced by, the *osmotic pressure* outside.

- » An *isotonic* solution has the same concentration of solute and solvent as found inside a cell, so a cell placed in isotonic solution — typically 1 percent saline solution for humans — experiences equal flow of water into and out of the cell, maintaining equilibrium.
- » A *hypotonic* solution has a lower solute concentration than that of the cell, which means the area with the highest osmolarity (less water) is inside the cell. If a human cell is placed in a hypotonic solution, such as distilled water, for example, water molecules move down the concentration gradient into the cell. This causes the cell to swell and could continue until the cell membrane bursts.
- » A *hypertonic* solution has a higher osmolarity outside of the cell. So the membrane of a human cell placed in 10 percent saline solution (10 percent salt and 90 percent water) would let water flow out of the cell (to where osmotic pressure is lower because water concentration is lower), therefore causing it to shrivel.

Active transport

Often, our bodies want to maintain an imbalance of concentration inside versus outside of a cell. This is particularly true for ions: Their imbalance is the foundation of nervous physiology, and our ability to move these ions allows us to influence osmosis. Since ions, for example, are being pumped against their concentration gradient, this movement requires energy. *Active transport* occurs across a semipermeable membrane, moving from the area of lower concentration to the area of higher concentration. Active transport also lets cells obtain nutrients that can't pass through the membrane by other means. Embedded in the cell membrane are transmembrane proteins able to detect and move compounds through the membrane. These *carrier* or *transport* proteins use ATP to pump molecules against their concentration gradients.

Some molecules (many proteins, for example) are far too large to be transported through the cell membrane; the hole it would need to move through would cause the cytoplasm to spill out. Bringing these molecules in, *endocytosis*, or out, *exocytosis*, requires energy, but concentration gradient is irrelevant for this type of active transport. Large molecules such as hormones are manufactured in a cell but must be released in order to carry out their function. The product (hormone) is packaged in an organelle called a *vesicle*.

The outer wall of the vesicle is a phospholipid bilayer, just like the cell membrane. When the vesicle reaches the cell membrane, its phospholipids align with those of the membrane. Thus, the contents of the vesicle have been released into the extracellular fluid without ever “opening” the cell.

Endocytosis occurs in much the same way, just in reverse. During *pinocytosis*, a fluid droplet is pulled into the cell as the membrane pinches in. (Again, phospholipids realign never exposing the inside of the cell to the outside environment.) *Phagocytosis* is similar to pinocytosis but occurs with solid particles. When the particle makes contact with the cell, the neighboring phospholipids project outward, engulfing the particle. In order to target specific molecules, such as cholesterol, cells utilize *receptor-mediated endocytosis*. Here, the targeted molecule binds to the receptor and the membrane extends around them both, bringing them into the cell.

6-9 Fill in the blanks to complete the following sentences:

The lipid bilayer structure of the cell membrane is made possible because phospholipid molecules contain two distinct regions: The

6. _____ region is attracted to water. The **7.** _____ of the phospholipids thus create an internal and external membrane surface. The **8.** _____ region is repelled by water, so the **9.** _____ of the phospholipids create an internal layer that will not interact with water.

10 What is a hypotonic solution?

- a. A solution that has a lower concentration of water than exists in the cell
- b. A solution that has a greater concentration of water than exists in the cell
- c. A solution that has a lower concentration of solute inside the cell
- d. A solution that has a greater concentration of solute outside the cell
- e. A solution that constantly varies in concentration when compared to what exists in the cell

11 Injecting a large quantity of distilled water into a human’s veins would cause many red blood cells to

- a. swell and burst.

- b. shrink or shrivel.
- c. carry more oxygen.
- d. aggregate.
- e. remain as normal.

12 Which of the following processes requires energy from ATP?

- I. diffusion/osmosis
- II. endo/exocytosis
- III. facilitated diffusion

- a. I and II
- b. II and III
- c. II only
- d. III only
- e. all require energy

13-16 Match the transport term to the description.

- a. active transport
- b. exocytosis
- c. pinocytosis
- d. phagocytosis

13. _____ Droplets of liquid are brought into the cell.

14. _____ Cell membrane extends around a molecule engulfing it into the cell.

15. _____ Molecules are pumped from areas of low concentration to high concentration.

16. _____ Molecules are released from the cell by a vesicle fusing with the cell membrane.

17 Transcytosis occurs in layers of cells that form barriers. One end of a cell

has receptors to bind a specific molecule. The cell brings that molecule in but then immediately sends it out on the other side. Explain the two processes that make up transcytosis.

Aiming for the Nucleus

The cell nucleus is the largest cellular organelle and the first to be discovered by scientists. On average, it accounts for about 10 percent of the total volume of the cell, and it holds a complete set of genes.

The outermost part of this organelle is the *nuclear envelope*, which is composed of a double-membrane barrier, each membrane of which is made up of a phospholipid bilayer. Between the two membranes is a fluid-filled space called the *perinuclear space*. The two layers fuse to form a selectively permeable barrier, but large *pores* allow relatively free movement of molecules and ions, including large proteins. Intermediate filaments lining the surface of the nuclear envelope make up the *nuclear lamina*, which functions in the disassembly and reassembly of the nuclear membrane during mitosis and binds the membrane to the endoplasmic reticulum. The nucleus also contains *nucleoplasm*, a clear viscous material that forms the matrix in which the subnuclear bodies are embedded.

DNA is packaged inside the nucleus in linear structures called *chromatin*. During cell division (see [Chapter 4](#)), the chromatin condenses and becomes coiled, forming *chromosomes*. Each chromosome is a DNA molecule encoded with the genetic information needed to direct the cell's activities. The most prominent subnuclear body is the *nucleolus*, a small spherical body that stores RNA molecules and produces *ribosomes*, which are exported to the cytoplasm where they translate *messenger RNA* (mRNA) to begin protein synthesis.

The following is an example question about the nucleus:



EXAMPLE Q. Identify the only cellular organelle found within the nucleus.

- a. lamina
- b. envelope
- c. nucleolus
- d. chromosome

A. The correct answer is nucleolus. The other options aren't organelles. True, chromosomes are found inside the cell's nucleus, but they don't serve the cell as an organ might serve the body. Remember: The suffix “*-elle*” is a diminutive that makes the word *organelle* translate into “little organ.”

18 What is the role of the nuclear lamina?

- a. to create a fluid-filled space within the nuclear envelope
- b. to assemble and store ribosomes
- c. a matrix to support the nuclear contents
- d. a protein packaging system within the cell
- e. to break down and rebuild the envelope

19 In the nucleus of a cell that isn't dividing, DNA

- a. moves about freely.
- b. is packaged within chromatin threads.
- c. regulates the activity of mitochondria.
- d. is organized into chromosomes.

20 The nucleolus

- a. packages DNA.
- b. enables large molecule transport.

- c. forms a membrane around the nucleus.
- d. assembles ribosomes.
- e. supports the contents of the nucleus.

Looking Inside: Organelles and Their Functions

Molecules that pass muster with the cell membrane enter the *cytoplasm*, a mixture of macromolecules (such as proteins and RNA), small organic molecules (such as glucose), ions, and water. Because of the various materials in the cytoplasm, it's a *colloid*, or mixture of phases, that alternates from a *sol* (a liquid colloid with solid suspended in it) to a *gel* (a colloid in which the dispersed phase combines with the medium to form a semisolid material). The fluid part of the cytoplasm, called the *cytosol*, has a differing consistency based on changes in temperature, molecular concentrations, pH, pressure, and agitation.

Within the cytoplasm lies a network of fibrous proteins collectively referred to as the *cytoskeleton*. It's not rigid or permanent, but changing and shifting according to the activity of the cell. The cytoskeleton maintains the cell's shape, enables it to move, anchors its organelles, and directs the flow of the cytoplasm. The fibrous proteins that make up the cytoskeleton include the following:

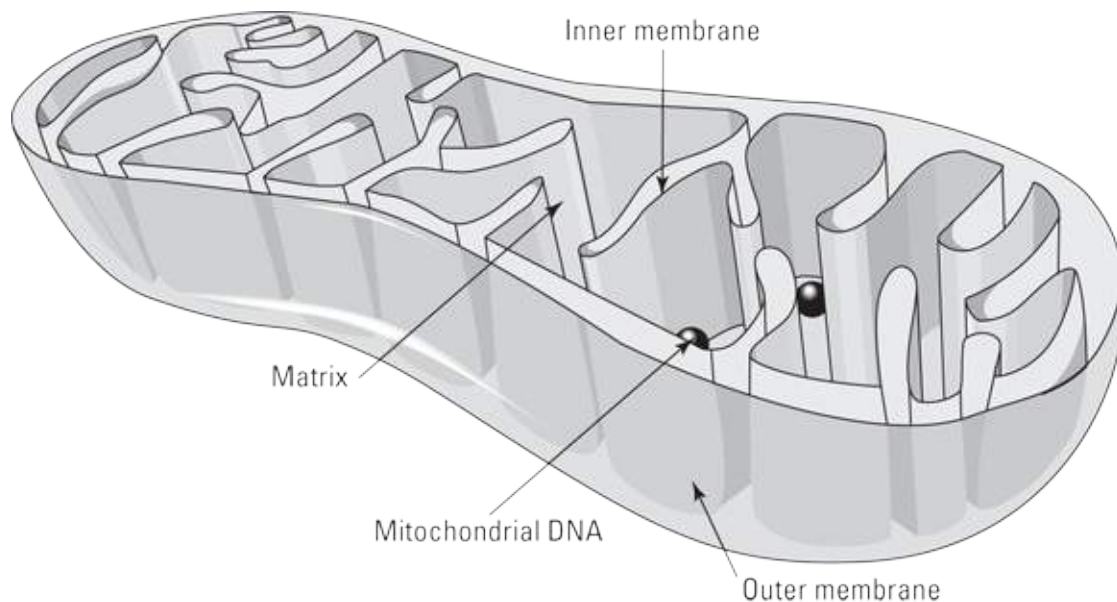
- » *Microfilaments*, rodlike structures about 5 to 8 nanometers wide that consist of a stacked protein called *actin*, the most abundant protein in eukaryotic cells. They provide structural support and have a role in cell and organelle movement as well as in cell division.
- » *Intermediate filaments*, the strongest and most stable part of the cytoskeleton. They average about 10 nanometers wide and consist of interlocking proteins, including *keratin*, that are involved in maintaining cell integrity and resisting forces that pull on the cell.
- » Hollow *microtubules* about 25 nanometers in diameter and 25 to 50 microns long made of the protein *tubulin* that grow with one end embedded in the *centrosome* near the cell's nucleus. Like microfilaments,

these components of cilia, flagella, and centrioles provide structural support, can shorten and elongate, and have a role in cell and organelle movement as well as in cell division.

Organelles, literally translated as “little organs,” are nestled inside the cytoplasm (except for the two organelles that move, *cilia* and *flagellum*, which are found on the cell’s exterior). Each organelle has different responsibilities for producing materials used elsewhere in the cell or body. Here are the key organelles and what they do:

- » **Centrosome:** Microtubules sprout from this structure, which is located next to the nucleus and is composed of two *centrioles* — arrays of microtubules — that function in separating genetic material during cell division.
- » **Cilia:** These are short, hairlike cytoplasmic projections on the external surface of the cell. In multicellular animals, including humans, cilia move materials over the surface of the cell. In some single-celled organisms, they’re used for locomotion.
- » **Endoplasmic reticulum (ER):** The ER is the site of lipid and protein synthesis. This organelle makes direct contact with the cell nucleus and is composed of membrane-bound canals and cavities that aid in the transport of materials throughout the cell. The two types of ER are *rough*, which is dotted with ribosomes on the outer surface, and *smooth*, which have no ribosomes on the surface. The rough ER manufactures membranes and secretory proteins in collaboration with the ribosomes. The smooth ER has a wide range of functions, including carbohydrate and lipid synthesis as well as breaking down drugs.
- » **Flagellum:** This whiplike cytoplasmic projection lies on the cell’s exterior surface. Found in humans primarily on sperm cells where it’s used for locomotion.
- » **Golgi apparatus (or body):** This organelle consists of a stack of flattened sacs with membranes. *Transport vesicles* carry newly made proteins from the ER to the golgi, and *secretory vesicles* carry the golgi’s products to the cell membrane. Located near the nucleus, it functions in the storage, modification, and packaging of proteins for secretion to various destinations within the cell or for release.

- » **Lysosome:** A tiny, membranous sac containing acids and digestive enzymes, the lysosome breaks down large molecules such as proteins, carbohydrates, and nucleic acids into materials that the cell can use. The enzymes can be used to open vesicles (break down their lipid bilayer) to access material brought in via endocytosis. It also destroys foreign particles, including bacteria and viruses, and helps to remove nonfunctioning structures from the cell.
- » **Mitochondrion:** Called the “powerhouse of the cell,” this usually rod-shaped organelle consists of two membranes — a smooth outer membrane and an invaginated (folded inward) inner membrane that divides the organelle into compartments. The inward folds of the inner membrane are called *cristae*. (See [Figure 3-2](#).) The mitochondrion provides critical functions in cell respiration, which we cover in [Chapter 2](#).
- » **Ribosomes:** These roughly 25-nanometer structures may be found along the endoplasmic reticulum or floating free in the cytoplasm. Composed of 60 percent RNA and 40 percent protein, they translate the genetic information on messenger RNA molecules to *synthesize*, or produce, a protein molecule.



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FIGURE 3-2: A mighty mitochondrion.

Answer these practice questions about cell organelles:

21 What is the function of a mitochondrion?

- a. To modify proteins, lipids, and other macromolecules
- b. To produce energy through aerobic respiration
- c. To keep the cell clean and recycle materials within it
- d. To produce and export hormones
- e. To assemble amino acids into proteins

22 Actin, which is the most abundant protein in human cells,

- a. forms filaments that contribute to the cytoskeleton.
- b. metabolizes lipids for cell energy.
- c. prevents binding of microtubules.
- d. synthesizes hormones.
- e. provides the cytoplasm its fluidity.

23 Which organelle translates genetic information to direct protein synthesis?

- a. The ribosome
- b. The lysosome
- c. The centriole
- d. The nucleus
- e. The vesicle

24 Which organelle is the site of protein synthesis?

- a. Smooth (agranular) endoplasmic reticulum
- b. Golgi apparatus
- c. Nucleus
- d. Rough (granular) endoplasmic reticulum
- e. Mitochondria

25 Which of the following can change the consistency of cytoplasm?

- a. Changes in pH
- b. Temperature
- c. Pressure
- d. Molecule concentration
- e. All of the above

26-38 Use the terms that follow to identify the cell structures and organelles shown in [Figure 3-3](#).

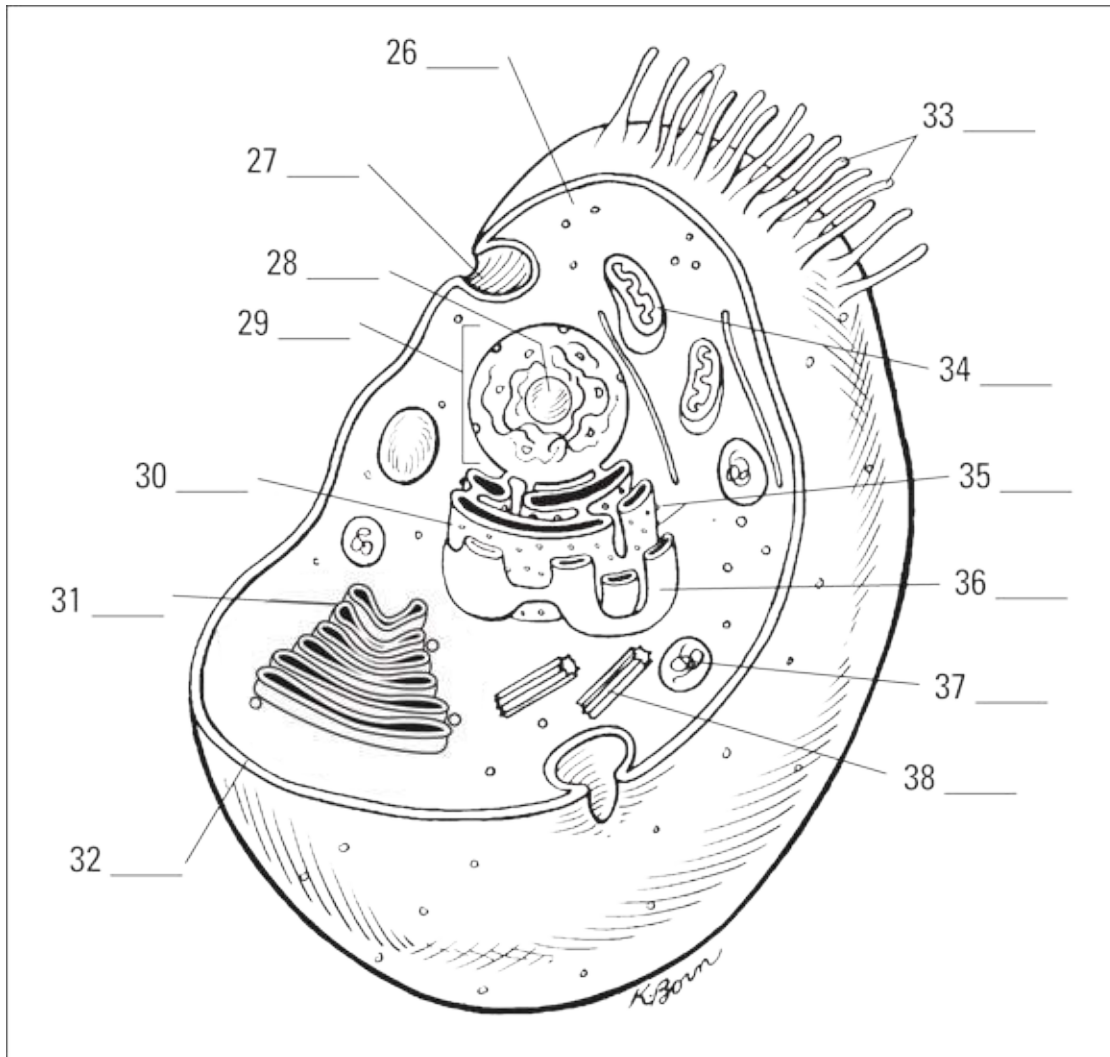


Illustration by Kathryn Born, MA

FIGURE 3-3: A cutaway view of an animal cell and its organelles.

- a. Centriole
- b. Cilia
- c. Cytoplasm
- d. Golgi apparatus
- e. Lysosome
- f. Mitochondrion
- g. Nucleolus
- h. Nucleus
- i. Cell (plasma) membrane
- j. Ribosomes
- k. Rough endoplasmic reticulum
- l. Smooth endoplasmic reticulum
- m. Vesicle (formation)

39-44 Match the organelles with their descriptions.

- a. Long, whiplike organelle for locomotion
 - b. Fluidlike interior of the cell that may become a semisolid, or colloid
 - c. Membranous sacs containing digestive enzymes
 - d. Site of cellular respiration
 - e. Site of ribosome production
 - f. Structure used for transporting molecules in, out, or around the cell
-
- 39. _____ Mitochondrion
 - 40. _____ Nucleolus
 - 41. _____ Flagellum
 - 42. _____ Cytoplasm
 - 43. _____ Lysosomes
 - 44. _____ Vesicle

Putting Together New Proteins

Proteins are essential building blocks for all living systems, which helps explain why the word is derived from the Greek term *proteios*, meaning “holding first place.” Cells use proteins to perform a variety of functions, including providing structural support and catalyzing reactions. Cells synthesize proteins through a systematic procedure that begins in the nucleus when the genetic code for a certain protein is *transcribed* from the cell’s DNA into *messenger RNA*, or *mRNA*. The mRNA moves through nuclear pores to the rough endoplasmic reticulum (RER), where ribosomes *translate* the message one *codon* of three nucleotides, or *base pairs*, at a time. The ribosome uses *transfer RNA*, or *tRNA*, to fetch each required amino acid from the cytoplasm and then link them together through *peptide bonds* to form proteins (see [Figure 3-4](#) for details).



FIGURE 3-4: The process of protein synthesis.

The free ribosomes are responsible for production of proteins that remain in the cell. Proteins to be sent out are synthesized by ribosomes attached to the ER. Regardless, the linking of amino acids into an incredibly long chain does